Supplementary methods and configuration

This note restates the configuration in ews.py and sections 2 and 3 of the manuscript. It is not a second paper and it does not report a new result.

Algorithm

KG-EWS takes noisy measurements, the committee predicates, and the fit, validation, and test indexes. It returns test scores and an alert threshold.

1. Fit the logistic model and, separately, the gradient-boosting baseline on the fit slice.
2. Compute evidence as the weighted sum of the eight rule indicators.
3. For each evidence weight on the grid, form the score on the validation slice and choose the threshold that minimizes decision cost.
4. Keep the weight with the smallest validation cost, and freeze that weight and its threshold.
5. Return the test scores, the alerts, and the names of the rules that fired.

The score is the logistic function of the logistic log-odds plus the evidence weight times the evidence. If the rules do not help, the chosen weight is zero and the score is the logistic score. The evidence is not added to the gradient-boosting score.

Rules

R1 through R7 are risk rules. P1 is protective. They fire only when every part of the antecedent holds on the clean operating state. The expert weights are 1.00, 0.85, 0.80, 0.95, 0.60, 0.55, 0.75, and −1.05. They are not the coefficients in Equation (4).

Models

Piece	Setting in ews.py
Logistic regression	class_weight balanced, max_iter 800, StandardScaler fit on the fit slice
Knowledge base	Logistic regression on the one-dimensional evidence, max_iter 500, not class-balanced
Gradient boosting	180 trees, learning rate 0.05, max depth 3, subsample 0.90, min_samples_leaf 12, random_state 42
Evidence-weight grid	16 values from 0 to 3 inclusive, step 0.2
Primary miss weight	5, with false-alarm weight 1
Sensitivity	Miss weights 3 and 8. The weight and threshold are chosen again. The models are not refit.
Stress	60 cohorts, seeds 12000–12059. Each of the eight predicate bits flips independently with probability 0.15.

What the manuscript claims, and what it does not

On the 120 primary cohorts, KG-EWS has the lowest mean decision cost (0.598) and the highest mean AUC (0.765) of the four scores. The paired cost advantage versus logistic regression is 0.024 (59 cohorts cheaper, 28 more expensive, 33 tied). The advantage remains at miss weight 3 and is not detectable at miss weight 8. It is absent in the predicate-flip block.

The thresholds of the rules are the same thresholds used to generate failures. Expert weights differ from the generating coefficients. There is no failure date. The five-to-one cost is a stated preference. Those limits are in sections 5 and 6 of the manuscript.

Reproduction

Repository: https://github.com/divzmunagala03-tech/Dba

Command, from the repository root: python3 ews.py

Libraries are pinned in 03_Supporting_Data/requirements.txt. Primary seeds are 9000 through 9119.